School of Computer Science and Engineering
Experiment List for Programming Ability and Logic Building - 1

Proposed Date	Lecture	Experiment	In Class / Take Home
2/26) Week: (02/02/26 to 07/0	1	Given an array **arr[]** of positive integers, where each value represents the number of chocolates in a packet. Each packet can have a variable number of chocolates. There are **m** students, the task is to distribute chocolate packets among **m** students such that - i. Each student gets **exactly** one packet. ii. The difference between maximum number of chocolates given to a student and minimum number of chocolates given to a student is minimum and return that minimum possible difference. **Examples:** **Input:** arr = [3, 4, 1, 9, 56, 7, 9, 12], m = 5 **Output:** 6 **Explanation:** The minimum difference between maximum chocolates and minimum chocolates is 9 - 3 = 6 by choosing following m packets :[3, 4, 9, 7, 9]. **Input:** arr = [7, 3, 2, 4, 9, 12, 56], m = 3 **Output:** 2 **Explanation:** The minimum difference between maximum chocolates and minimum chocolates is 4 - 2 = 2 by choosing following m packets :[3, 2, 4]. **Input:** arr = [3, 4, 1, 9, 56], m = 5 **Output:** 55 **Explanation**: With 5 packets for 5 students, each student will receive one packet, so the difference is 56 - 1 = 55. Link: https://www.geeksforgeeks.org/problems/chocolate-distribution-problem3825/1	In Class
	1	Given a number **x** and an array of integers **arr**, find the smallest subarray with sum greater than the given value. If such a subarray do not exist return 0 in that case. **Examples:** **Input:** x = 51, arr[] = [1, 4, 45, 6, 0, 19] **Output:** 3 **Explanation:** Minimum length subarray is [4, 45, 6] **Input:** x = 100, arr[] = [1, 10, 5, 2, 7] **Output:** 0 **Explanation:** No subarray exist Link: https://www.geeksforgeeks.org/problems/smallest-subarray-with-sum-greater than-x5651/1	In Class

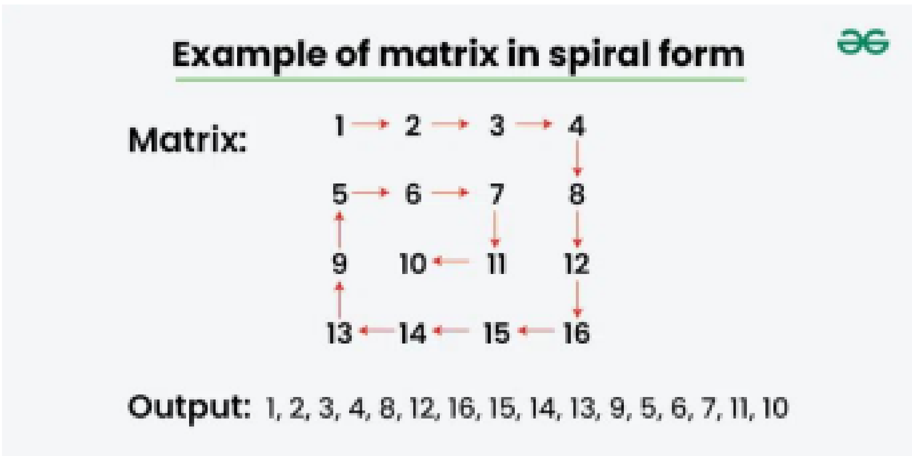
	1	Given an **array** and a range **a**, **b**. The task is to partition the array around the range such that the array is divided into three parts. 1) All elements smaller than **a** come first. 2) All elements in range **a** to **b** come next. 3) All elements greater than **b** appear in the end. The individual elements of three sets can appear in any order. You are required to return the modified array. **Note**: The generated output is true if you modify the given array successfully. Otherwise false. **Geeky Challenge:** Solve this problem in O(n) time complexity. **Examples:** **Input**: arr[] = [1, 2, 3, 3, 4], a = 1, b = 2	Take Home

Proposed Date	Lecture	Experiment	In Class / Take Home
		Output: true **Explanation**: One possible arrangement is: {1, 2, 3, 3, 4}. If you return a valid arrangement, output will be true. **Input**: arr[] = [1, 4, 3, 6, 2, 1], a = 1, b = 3 **Output**: true **Explanation:** One possible arrangement is: {1, 3, 2, 1, 4, 6}. If you return a valid arrangement, output will be true. Link: https://www.geeksforgeeks.org/problems/three-way-partitioning/1	

	1	Given an array **arr** and a number **k**. One can apply a swap operation on the array any number of times, i.e choose any two index i and j (i < j) and swap arr[i] , arr[j] . Find the **minimum** number of swaps required to bring all the numbers less than or equal to **k** together, i.e. make them a contiguous subarray. **Examples :** **Input:** arr[] = [2, 1, 5, 6, 3], k = 3 **Output:** 1 **Explanation:** To bring elements 2, 1, 3 together, swap index 2 with 4 (0-based indexing), i.e. element arr[2] = 5 with arr[4] = 3 such that final array will be- arr[] = [2, 1, 3, 6, 5] **Input:** arr[] = [2, 7, 9, 5, 8, 7, 4], k = 6 **Output:** 2 **Explanation:** To bring elements 2, 5, 4 together, swap index 0 with 2 (0-based indexing) and index 4 with 6 (0-based indexing) such that final array will be- arr[] = [9, 7, 2, 5, 4, 7, 8] **Input:** arr[] = [2, 4, 5, 3, 6, 1, 8], k = 6 **Output:** 0 Link: https://www.geeksforgeeks.org/problems/minimum-swaps-required-to-bring-all elements-less-than-or-equal-to-k-together4847/1	Take Home
	1	Given an array **arr[]** of positive integers. Return true if all the array elements are palindrome otherwise, return false. **Examples:** **Input:** arr[] = [111, 222, 333, 444, 555] **Output:** true **Explanation:** arr[0] = 111, which is a palindrome number. arr[1] = 222, which is a palindrome number. arr[2] = 333, which is a palindrome number. arr[3] = 444, which is a palindrome number. arr[4] = 555, which is a palindrome number. As all numbers are palindrome so This will return true. **Input:** arr[] = [121, 131, 20] **Output:** false **Explanation:** 20 is not a palindrome hence the output is false. Link:	Take Home

School of Computer Science and Engineering
Experiment List for Programming Ability and Logic Building - 1

Proposed Date	Lect	Experiment	In Class / Take Home

	u r e		
2/26) Week: (02/02/26 to 07/0	2	Given an array **arr[]** of integers, calculate the median. **Examples:** **Input:** arr[] = [90, 100, 78, 89, 67] **Output:** 89 **Explanation:** After sorting the array middle element is the median **Input:** arr[] = [56, 67, 30, 79] **Output:** 61.5 **Explanation:** In case of even number of elements, average of two middle elements is the median. **Input:** arr[] = [1, 2] **Output:** 1.5 **Explanation:** The average of both elements will result in 1.5. Link: https://www.geeksforgeeks.org/problems/find-the-median0527/1	In Class
	2	You are given a rectangular matrix **mat[][]** of size **n x m**, and your task is to return an array while **traversing** the matrix in **spiral** form. **Examples:** **Input**: mat[][] = [[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12], [13, 14, 15, 16]] **Output**: [1, 2, 3, 4, 8, 12, 16, 15, 14, 13, 9, 5, 6, 7, 11, 10] **Explanation**:  **Input**: mat[][] = [[1, 2, 3, 4, 5, 6], [7, 8, 9, 10, 11, 12], [13, 14, 15, 16, 17, 18]] **Output**: [1, 2, 3, 4, 5, 6, 12, 18, 17, 16, 15, 14, 13, 7, 8, 9, 10, 11] **Explanation**: Applying same technique as shown above. **Input**: mat[][] = [[32, 44, 27, 23], [54, 28, 50, 62]] **Output**: [32, 44, 27, 23, 62, 50, 28, 54] **Explanation**: Applying same technique as shown above, output will be [32, 44, 27, 23, 62, 50, 28, 54]. Link: https://www.geeksforgeeks.org/problems/spirally-traversing-a-matrix 1587115621/1	In Class
	2	You are given an m x n integer matrix matrix with the following two	Take

		properties: • Each row is sorted in non-decreasing order.	Home

School of Computer Science and Engineering
Experiment List for Programming Ability and Logic Building - 1

Proposed Date	Lecture	Experiment	In Class / Take Home

• The first integer of each row is greater than the last integer of the previous row. Given an integer target, return true *if* target *is in* matrix *or* false *otherwise*. You must write a solution in O(log(m * n)) time complexity.

Example 1:

1	3	5	7
10	11	16	20
23	30	34	60

Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 3
Output: true
Example 2:

1	3	5	7
10	11	16	20
23	30	34	60

Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 13
Output: false

Link: https://leetcode.com/problems/search-a-2d-matrix/description/

2 Given a **row-wise sorted** matrix **mat[][]** of size n*m, where the number of rows and columns is always **odd**. Return the **median** of the matrix. Take Home
Examples:
Input: mat[][] = [[1, 3, 5],
[2, 6, 9],
[3, 6, 9]]
Output: 5

School of Computer Science and Engineering
Experiment List for Programming Ability and Logic Building - 1

Proposed Date	Lecture	Experiment	In Class / Take Home
		Explanation: Sorting matrix elements gives us [1, 2, 3, 3, 5, 6, 6, 9, 9]. Hence, 5 is median. **Input:** mat[][] = [[2, 4, 9], [3, 6, 7], [4, 7, 10]] **Output:** 6 **Explanation**: Sorting matrix elements gives us [2, 3, 4, 4, 6, 7, 7, 9, 10]. Hence, 6 is median. **Input:** mat = [[3], [4], [8]] **Output:** 4 **Explanation**: Sorting matrix elements gives us [3, 4, 8]. Hence, 4 is median. Link: [https://www.geeksforgeeks.org/problems/median-in-a-row-wise-sorted matrix1527/1](https://www.geeksforgeeks.org/problems/median-in-a-row-wise-sorted)	
	2	You are given a 2D binary array **arr[][]** consisting of only 1s and 0s. Each row of the array is sorted in non-decreasing order. Your task is to find and return the index of the first row that contains the maximum number of 1s. If no such row exists, return -1. **Note:** • The array follows 0-based indexing. • The number of rows and columns in the array are denoted by n and m respectively. **Examples:** **Input:** arr[][] = [[0,1,1,1], [0,0,1,1], [1,1,1,1], [0,0,0,0]] **Output:** 2 **Explanation:** Row 2 contains the most number of 1s (4 1s). Hence, the output is 2. **Input:** arr[][] = [[0,0], [1,1]] **Output:** 1 **Explanation:** Row 1 contains the most number of 1s (2 1s). Hence, the output is 1. **Input:** arr[][] = [[0,0], [0,0]] **Output:** -1 **Explanation:** No row contains any 1s, so the output is -1. Link: https://www.geeksforgeeks.org/problems/row-with-max-1s0023/1	Take Home